Why Lasso Creates Sparsity

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Where this fits: Pipeline map step 8 (Model). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	**8 Model**	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Lasso regression.
- **Compare with:** Ridge regression; L1 and L2 regularisation in neural networks.

1. Overview

Key point: The Lasso penalty puts a sharp corner into the loss curve at **slope** (G-1823) 0, and a large enough λ makes that corner the lowest point. The Ridge penalty keeps the curve smooth, so its lowest point only moves towards 0.

One feature, the slope reaches exactly 0 showed that **Lasso regression** (G-1047) sets coefficients to exactly 0, while **Ridge regression** (G-1691) only shrinks them. A model in which many coefficients are exactly 0 is called **sparse** (G-1846), so the effect is called **sparsity** (G-1849).

Sparsity matters when many **features** (G-772) (input variables, one column of the data table each) are useless for the prediction. Lasso removes those features from the equation, which leaves a simpler model that is easier to read. Ridge keeps every feature, so it suits data in which most features are useful.

"Why does Lasso create sparsity, and Ridge does not?" is one of the most common interview questions on **regularisation** (G-1659). This Note answers it for one feature, in this order:

- the picture: the loss curve and its corner (section 2);
- the formula for the Lasso slope, in cases (section 3);
- the numbers: the slope reaching 0 and staying there (section 4);
- the same effect for many features (section 5);
- a check against scikit-learn (section 6).

2. The picture: a corner in the loss curve

Key point: The best **slope** (G-1823) is the lowest point of the loss curve. Lasso's penalty $\lambda|m|$ bends the curve into a corner at $m = 0$; once λ is large enough, the corner is the lowest point.

2.1 The best slope is the lowest point of a curve

Take one **feature** (G-772) x and a **target** (G-1949) y (the output we predict). Every **slope** (G-1823) m gives a different line, and every line has a total squared error, the **residual sum of squares** (G-1684). Plotting that error against the slope gives a U-shaped curve, a parabola. The best slope is the bottom of the U.

A three-point toy set shows each quantity with numbers. Here $n = 3$ is the number of **observations** (G-1374), x_i the feature of the i-th one and y_i its target:

i	x_i	y_i
1	1	1
2	2	3
3	3	2

The bar means "mean". The means are:

$$\bar{x} = (1 + 2 + 3)/3 = 2$$

$$\bar{y} = (1 + 3 + 2)/3 = 2$$

To write the error curve down, two sums are enough. On the toy set, the first sum is:

$$(1 - 2)(1 - 2) + (2 - 2)(3 - 2) + (3 - 2)(2 - 2)$$

$$= 1 + 0 + 0 = 1$$

The second sum is:

$$(1 - 2)^2 + (2 - 2)^2 + (3 - 2)^2 = 1 + 0 + 1 = 2$$

In general they are called S and D. The symbol $\sum_{i=1}^{n}$ means "add the term for $i = 1$, then $i = 2$, up to $i = n$"; on the toy set $n = 3$, so each sum has the three terms written above.

$$S = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

$$D = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

- S measures how strongly x and y move together. S is positive when y tends to rise with x, and negative when it tends to fall.
- D measures how spread out x is. D is always positive.

So the toy set has $S = 1$ and $D = 2$. With the best **intercept** (G-960) $b = \bar{y} - m\bar{x}$ put in, the error of a point for slope m is $(y_i - \bar{y}) - m(x_i - \bar{x})$. Squaring one error, step by step (write $u = y_i - \bar{y}$ and $v = x_i - \bar{x}$):

$$(u - mv)^2 = u^2 - 2muv + m^2v^2$$

Adding over all the points, each of the three terms becomes a sum:

$$\sum u^2 = \sum (y_i - \bar{y})^2$$

$$\sum uv = S$$

$$\sum v^2 = D$$

so the total error is

$$\sum (y_i - \bar{y})^2 - 2Sm + Dm^2$$

Check on the toy set: the first sum is:

$$(1-2)^2 + (3-2)^2 + (2-2)^2 = 2$$

So the total error is $2 - 2m + 2m^2$. At $m = 0.5$ it is:

$$2 - 1 + 0.5 = 1.5$$

At $m = 1$ it is:

$$2 - 2 + 2 = 2$$

These are the same errors found by direct squaring: 2 for the line with slope 1 in the error function, and 1.5 for the best line, slope 0.5, in finding the minimum. In the notation of the curve:

$$\sum_{i=1}^{n} [(y_i - \bar{y}) - m(x_i - \bar{x})]^2 = Dm^2 - 2Sm + \text{constant}$$

The constant, $\sum (y_i - \bar{y})^2$, does not depend on m, so it moves the curve up or down without moving its lowest point. This Note uses $S = 100$ and $D = 50$ throughout. The curve $50m^2 - 200m$ has its bottom at $m = S/D = 2$, the linear regression slope.

2.2 Adding a penalty moves the lowest point

A **penalty term** (G-1476) is an extra cost for a large **slope** (G-1823), added to the squared error. Its strength is λ.

	Penalty	Loss curve (constant left out)
Ridge	λm^2	$Dm^2 - 2Sm + \lambda m^2$
Lasso	$2\lambda\|m\|$	$Dm^2 - 2Sm + 2\lambda\|m\|$

(The 2 in the Lasso penalty only rescales λ; section 3.1 explains it.)

Figure 1 draws both curves while λ grows from 0 to 150. Watch the black dot, the lowest point of each curve.

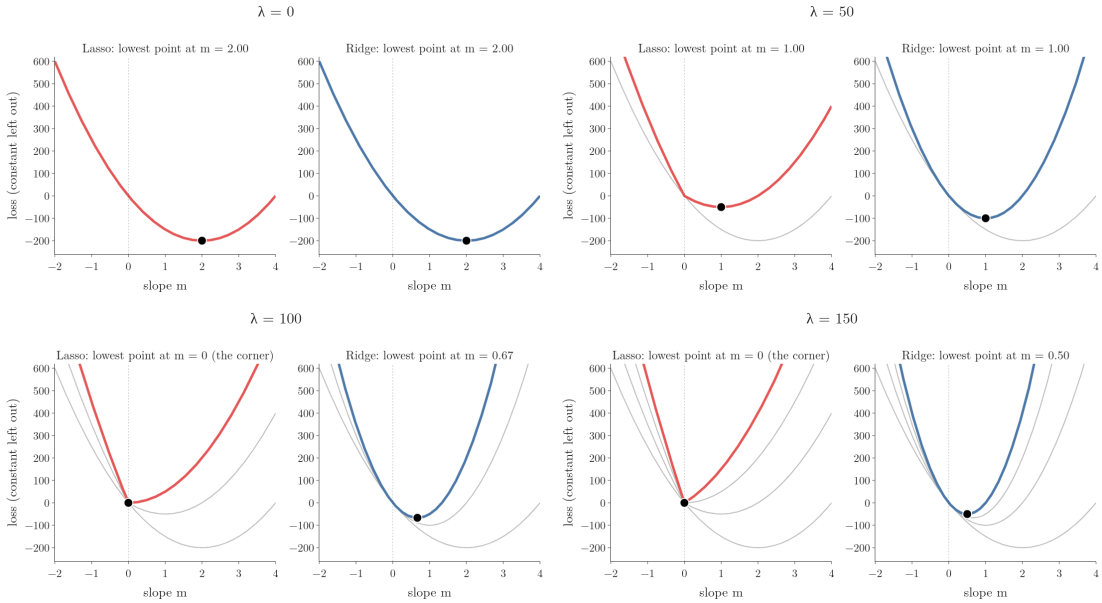


Figure 1: The one-feature loss against the slope m for S = 100 and D = 50, as λ grows from 0 to 150. Left: the Lasso curve grows a corner at m = 0, and from $\lambda = 100$ the lowest point is the corner itself. Right: the Ridge curve stays a smooth parabola, and its lowest point moves towards 0 without reaching it (m = 0.5 at $\lambda = 150$). Grey curves: $\lambda = 0$, 50 and 100. Idea after StatQuest, "Ridge vs Lasso Regression, Visualized!!!"; the curves are this Note's own numbers

Step by step, in Figure 1:

1. $\lambda = 0$. Both curves are the same parabola, with the lowest point at $m = 2$.
2. **Ridge, any λ.** Adding λm^2 to a parabola gives a narrower parabola. The curve is still smooth, and its bottom slides left: $m = 1$ at $\lambda = 50$, 0.67 at $\lambda = 100$, 0.5 at $\lambda = 150$. The bottom never lands on 0.

3. **Lasso, $\lambda = 50$.** The penalty $2\lambda|m|$ is a V with its point at $m = 0$. Adding a V to a parabola leaves a sharp corner in the curve at $m = 0$. The lowest point is still to the right of the corner, at $m = 1$.
4. **Lasso, $\lambda = 100$.** The corner has grown, and the lowest point has slid into it: $m = 0$.
5. **Lasso, $\lambda = 150$.** The curve now rises on both sides of the corner. The lowest point stays at $m = 0$.

A slope of exactly 0 means the **feature** (G-772) is no longer used in the prediction. The rest of the Note explains the same result with a formula: where the corner comes from, why the lowest point reaches it at $\lambda = 100$, and why it stays.

3. The Lasso slope

> **Key point:** Because of the **absolute value** (G-159), the derivative is different for m $>$ 0 and m $<$ 0, so the formula comes in cases.

Without a penalty, we set the **derivative** (G-595) of the curve of section 2.1 to zero (its **slope** (G-1823) is $2Dm - 2S$):

$$2Dm - 2S = 0$$

$$m = \frac{S}{D}$$

This is **linear regression**, with $b = \bar{y} - m\bar{x}$ (see the formulas of ordinary least squares). The Lasso slope is found the same way.

3.1 The loss

> **Key point:** The loss is the squared error plus $2\lambda|$m$|$; the 2 only keeps the result tidy.

With one **feature** (G-772), the Lasso loss is

$$L = \sum_{i=1}^{n} (y_i - mx_i - b)^2 + 2\lambda|m|$$

The factor 2 does not change the idea: it only rescales λ, and it makes the final formula simpler.

The penalty does not contain b, so differentiating with respect to b gives the OLS result again: $b = \bar{y} - m\bar{x}$. Substituting it, each error becomes $(y_i - \bar{y}) - m(x_i - \bar{x})$, as in section 2.1.

3.2 Why cases are needed

> **Key point:** |m| has no **derivative** (G-595) at m = 0, but on each side of 0 it is a simple line.

The **absolute value** (G-159) $|m|$ has a sharp corner at $m = 0$: the corner of the Lasso curve in Figure 1. A curve has no single **slope** (G-1823) at a corner, so $|m|$ is not **differentiable** (G-606) at $m = 0$. On either side, though, it is simple:

- if $m > 0$, then $|m| = m$;
- if $m < 0$, then $|m| = -m$.

So we solve the two cases separately.

3.3 Case m > 0

> **Key point:** For a positive **slope** (G-1823), m = (S − λ) / D.

With $|m| = m$, the penalty $2\lambda|m|$ is $2\lambda m$, whose slope is 2λ. The slope of the squared error comes from the chain rule, as in finding the minimum: each term gives $2 \times \text{error} \times (-(x_i - \bar{x}))$. Setting the total slope to zero (here $\partial L/\partial m$ means "how fast L changes when only m changes"):

$$\frac{\partial L}{\partial m} = -2\sum_{i=1}^{n}(x_i - \bar{x})\left[(y_i - \bar{y}) - m(x_i - \bar{x})\right] + 2\lambda = 0$$

Split the sum into the two sums S and D of section 2.1:

$$-2\left(S - mD\right) + 2\lambda = 0$$

$$-2S + 2mD + 2\lambda = 0$$

Move the terms without m to the right, then divide by $2D$:

$$2mD = 2S - 2\lambda$$

$$m = \frac{S - \lambda}{D}$$

Check with $S = 100$, $D = 50$, $\lambda = 50$: the equation is:

$$-200 + 100m + 100 = 0$$

So $m = 1$, and the formula gives:

$$(100 - 50)/50 = 1$$

3.4 Case m < 0

> **Key point:** For a negative **slope** (G-1823), m = (S + λ) / D.

With $|m| = -m$, the penalty is $-2\lambda m$, whose slope is -2λ instead of $+2\lambda$:

$$-2S + 2mD - 2\lambda = 0$$

$$2mD = 2S + 2\lambda$$

$$m = \frac{S + \lambda}{D}$$

Check with $S = -100$, $D = 50$, $\lambda = 50$: the equation is:

$$200 + 100m - 100 = 0$$

So $m = -1$, and the formula gives:

$$(-100 + 50)/50 = -1$$

3.5 Case m = 0

> **Key point:** When neither formula gives an answer with the right sign, the **slope** (G-1823) is 0.

Each case was solved assuming the sign of m, so each answer must be checked against that sign:

- Case 3.3 is valid only if its answer is positive, which needs $S > \lambda$.
- Case 3.4 is valid only if its answer is negative, which needs $S < -\lambda$.
- When S lies between $-\lambda$ and λ, neither case applies, and the lowest point of the loss is the corner itself: $m = 0$.

Condition	Lasso slope
$S > \lambda$	$m = (S - \lambda)/D$
$-\lambda \leq S \leq \lambda$	$m = 0$
$S < -\lambda$	$m = (S + \lambda)/D$

Figure 2 draws the **derivative** (G-595) of the loss, both cases, for $S = 100$, $D = 50$ and three values of λ. Watch the jump at the corner $m = 0$: once λ reaches S, the jump spans zero and the lowest loss sits on the corner.

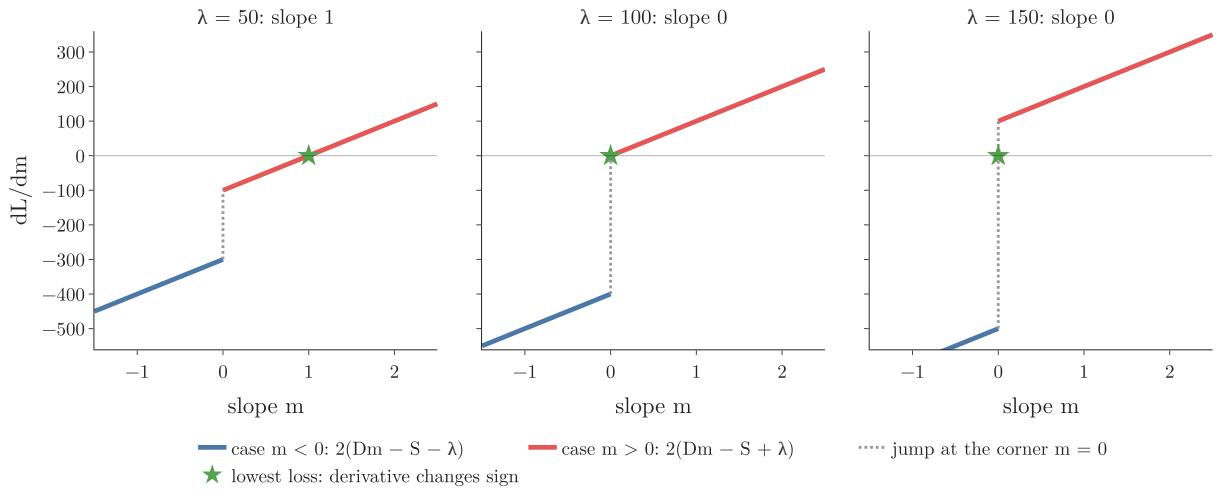


Figure 2: The derivative of the Lasso loss against the slope m (S = 100, D = 50). Blue: the m < 0 case; red: the m > 0 case; dotted: the jump at the corner. At $\lambda = 50$ the red line crosses zero at m = 1. At $\lambda = 100$ and 150 neither line crosses zero on its own side: the derivative jumps from negative to positive at m = 0, so the slope is 0.

3.6 The Ridge slope, for comparison

> **Key point:** Ridge: m = S / (D + λ). λ sits in the denominator.

The Ridge penalty λm^2 has no corner, so one **derivative** (G-595) covers every m (see ridge regression with one feature):

$$2Dm - 2S + 2\lambda m = 0$$

$$2m(D + \lambda) = 2S$$

$$m = \frac{S}{D + \lambda}$$

with the same b as before.

4. Watching the slope reach 0

> **Key point:** Subtracting λ from S brings the top of the fraction down to exactly 0. After that, the **slope** (G-1823) stays at 0.

Take the same numbers: $S = 100$ and $D = 50$, so the linear regression slope is:

$$100/50 = 2$$

Figure 3 follows both slopes as λ grows; the values are the lowest points of Figure 1.

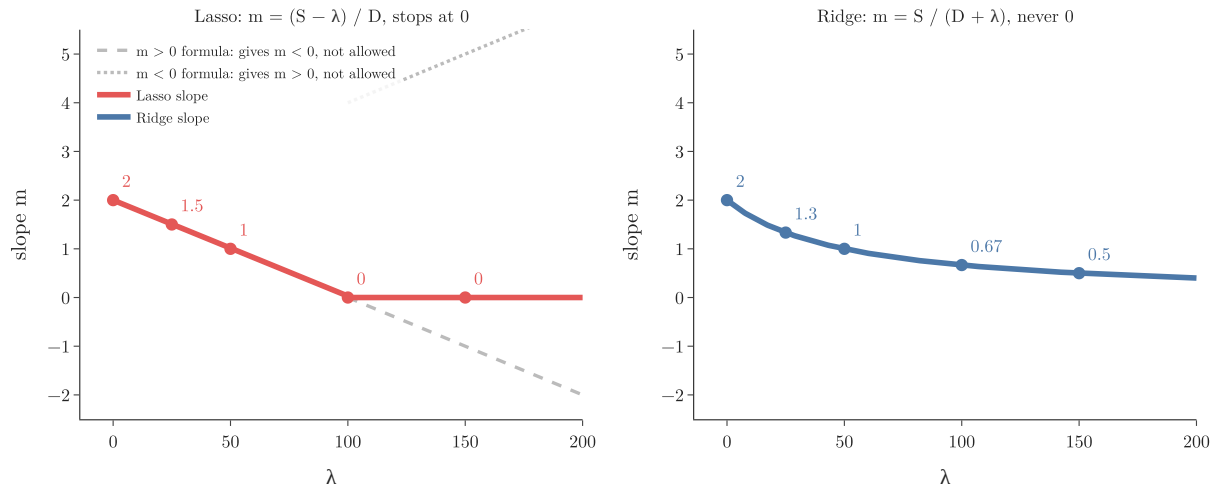


Figure 3: Slope against λ for Lasso and Ridge, with S = 100 and D = 50

λ	Lasso: $(100 - \lambda)/50$	Ridge: $100/(50 + \lambda)$
0	2	2
25	1.5	1.33
50	1	1
100	**0**	0.67
150	**0**	0.5
1000	**0**	0.095

4.1 Why Lasso reaches 0

> **Key point:** λ grows until it equals S; then the numerator is 0.

At $\lambda = 100$, the numerator $S - \lambda$ is 0, so the **slope** (G-1823) is exactly 0.

4.2 Why Lasso stops at 0

> **Key point:** Going past 0 would need the other case's formula, which pushes the **slope** (G-1823) back to the positive side.

Work through $\lambda = 150$ step by step:

1. The positive-case formula gives:
$$(100 - 150)/50 = -1$$
2. The value -1 is negative, so the positive-case formula no longer applies (Figure 3, dashed).
3. The negative-case formula gives:
$$(100 + 150)/50 = 5$$
4. The value 5 is positive, so the negative-case formula does not apply either (Figure 3, dotted).

Neither side has a valid answer, so the slope stays at $m = 0$. The same happens for every larger λ.

The mirror case. The same holds with a negative S. With $S = -100$ and $D = 50$, the negative-case formula $(S + \lambda)/D$ gives a slope of -2 at $\lambda = 0$, -1 at $\lambda = 50$ and 0 at $\lambda = 100$. At $\lambda = 150$:

1. The negative-case formula gives a positive value, so it does not apply:
$$(-100 + 150)/50 = 1$$
2. The positive-case formula gives a negative value, so it does not apply either:
$$(-100 - 150)/50 = -5$$

Again neither side has a valid answer, and the slope stays at 0.

4.3 Why Ridge never reaches 0

> **Key point:** With λ only in the denominator, the fraction gets smaller but its top never changes.

An everyday picture: take a bill of 100 rupees. Subtracting a discount of 100 rupees brings it to exactly 0. Splitting it among more and more people only makes each share smaller; no share ever becomes 0. Lasso subtracts λ; Ridge divides by something that grows with λ.

In Ridge's $S/(D + \lambda)$, the numerator stays at 100 however large λ gets. Even with $\lambda =$ 10,000,000, the **slope** (G-1823) is 100/10,000,050, tiny but not 0.

In short:

- **Lasso:** λ is in the **numerator**, so it can make the slope exactly 0.
- **Ridge:** λ is in the **denominator**, so it can only make the slope small.

5. The dead zone

> **Key point:** For a fixed λ, every feature whose S lies between $-\lambda$ and λ gets a **slope** (G-1823) of exactly 0.

Figure 4 turns the view around: λ is fixed at 100, and the slope is drawn for every value of S.

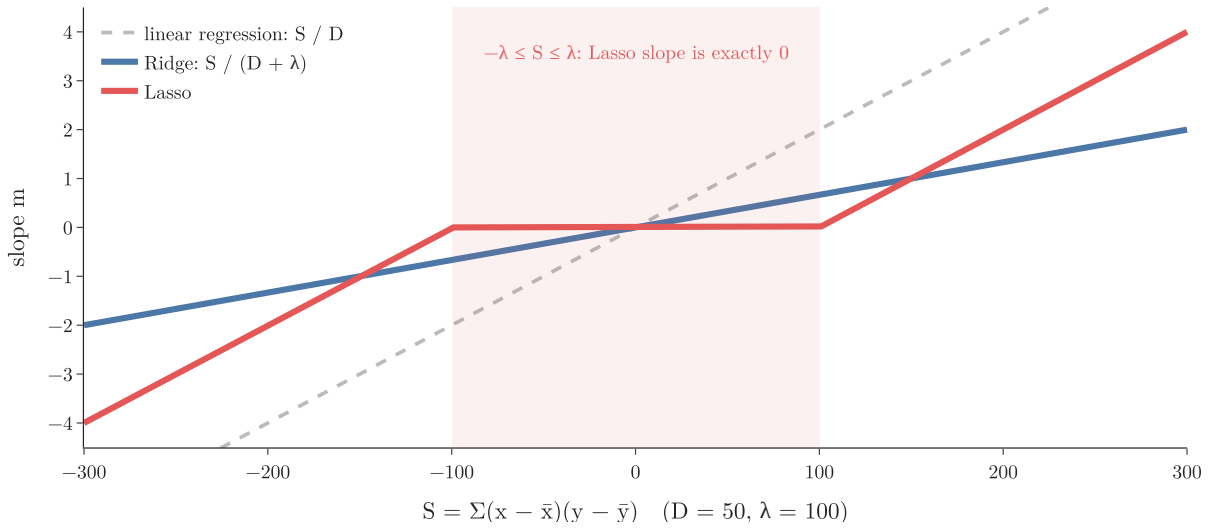


Figure 4: The slope against S for linear regression, Ridge and Lasso

- **Linear regression** (grey dashed): the slope is proportional to S.
- **Ridge** (blue): also proportional to S, just flatter. The Ridge slope is 0 only when S is exactly 0.
- **Lasso** (red): flat at 0 for every S between -100 and 100, the shaded **dead zone** (G-552). Outside it, Lasso follows the linear regression line moved $\lambda/D = 2$ towards 0.

S measures how strongly the feature and the **target** (G-1949) move together. So a feature with only a weak link to the target falls into the dead zone and is dropped. Dropping weak features in this way is Lasso's feature selection (G-768): the model that is left uses fewer **features** (G-772), so it is simpler and easier to read.

Think of λ as an entry fee. A feature's link with the target, $|S|$, must be larger than the fee to get any slope at all, and above the fee the feature keeps only what is left over.

> **Extra:** Shrinking a value towards 0 by a fixed amount and setting it to 0 if it would cross 0 is called **soft thresholding** (G-1827). With several features, Lasso has no single formula, but scikit-learn's method, **coordinate descent** (G-485), applies this same soft-threshold step to one coefficient at a time, over and over (ESL §3.8.6; scikit-learn docs, `Lasso`). The repeated soft-threshold step is why many coefficients land exactly on 0.

5.1 Ten features at once

> **Key point:** On the 10-feature diabetes data, Lasso coefficients drop to exactly 0 one after another; Ridge coefficients only shrink.

Figure 5 grows λ for Ridge (left) and Lasso (right) on the diabetes data of feature selection (same split). Each panel plots the coefficient (vertical axis) against λ (horizontal axis, log scale), one line per **feature** (G-772): bmi, bp, s1, s2 and s5 in colour, the other five in grey. Watch the dots at the end of the lines: a Lasso dot that reaches 0 turns into an open circle and stays there, while every Ridge dot keeps sliding towards 0 without arriving.

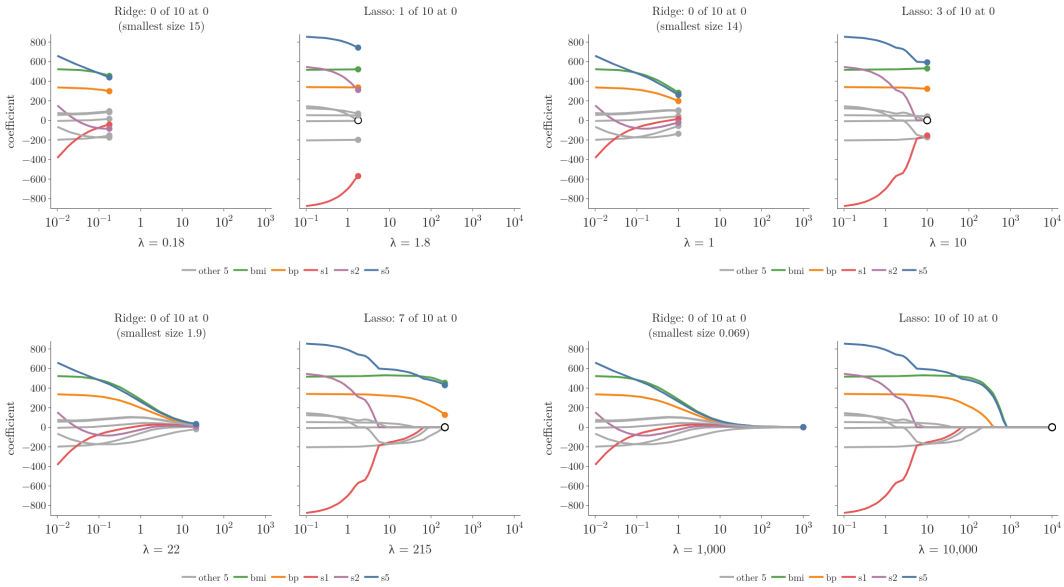


Figure 5: Ridge (left) and Lasso (right) coefficients of the 10 diabetes features as λ grows. Each panel has its own λ range, because Ridge shrinks faster on these features. An open circle marks a coefficient that is exactly 0

- **Lasso:** the coefficients reach 0 one at a time, bmi and s5 last; from $\lambda \approx 825$ all 10 are exactly 0.
- **Ridge:** at $\lambda = 1000$, all coefficients are tiny, but the smallest is still 0.069, not 0.

5.2 The picture: a diamond and a circle

> **Key point:** Lasso's feasible region is a diamond with corners on the axes; the loss rings around the least-squares answer usually touch it first at a corner, where one coefficient is exactly 0.

The same result has a geometric picture (ESL §3.4.3, Figure 3.11). Ridge and Lasso can each be written as "make the loss as small as possible while the coefficients stay inside a budget". The budget is a **constraint** (G-456), and the set of coefficients that meet it is the **feasible region** (G-759). With two coefficients and a budget t:

- **Lasso:** $|b_1| + |b_2| \leq t$, a diamond with its corners on the axes.
- **Ridge:** $b_1^2 + b_2^2 \leq t^2$, a circle.

The loss is a bowl over the (b_1, b_2) plane. Its lowest point is the **ordinary least squares** (G-1406) answer, and points of equal loss form ellipses around it, the loss **contours** (G-468; lines of equal loss, see reading a contour map). Growing the ellipse until it first touches the feasible region gives the answer: the point of the region with the smallest loss.

Figure 6 does this on two **features** (G-772) of the same diabetes training split, bmi and bp, each scaled to standard deviation 1, with the same budget $t = 15$ for both. Watch where each ring first meets its region.

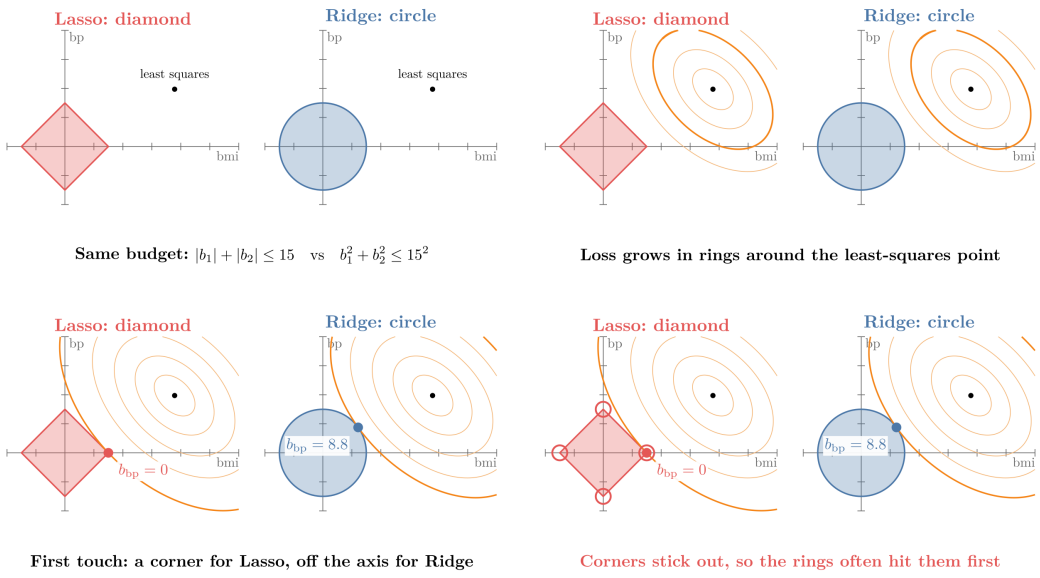


Figure 6: Lasso (left) and Ridge (right) with the same budget on two diabetes features. The loss ellipse grows from the least-squares point until it first touches each region: the diamond at its corner (bp coefficient exactly 0), the circle at a point where both coefficients are non-zero

- **Lasso:** the ring first touches the diamond exactly at its corner $(15, 0)$. The bp coefficient is exactly 0.
- **Ridge:** the circle has no corners, so the ring touches it at $(12.2, 8.8)$. Both coefficients are kept, just smaller than the least-squares $(37.9, 19.7)$.

The corners stick out towards the rings, so the rings often meet a corner first. With more features the diamond has many corners, edges and faces on which some coefficients are 0, so there are even more chances for exact zeros (ESL §3.4.3).

6. Checking with scikit-learn

> **Key point:** On the 100-observation example, the formula and scikit-learn's Lasso give the same **slopes** (G-1823), including the exact 0.

For the example of the earlier Notes, with 100 **observations** (G-1374) (records, one row of the data table each), the computer adds the two sums of section 2.1 over the 100 pairs (input: the 100 values of x and y; output: $S = 2416.73$ and $D = 86.85$). So the slope reaches 0 at $\lambda = 2416.73$. One row of the table by hand:

$$\frac{2416.73 - 500}{86.85} = \frac{1916.73}{86.85} = 22.07$$

λ	Formula	scikit-learn `Lasso`
500	22.071	22.071
1000	16.313	16.313
2000	4.799	4.799
2416.73	0	0
3000	0	0

Figure 7 checks many more values of λ. Watch the circles from scikit-learn sit on the formula's line, including the flat part after $\lambda = 2416.73$.

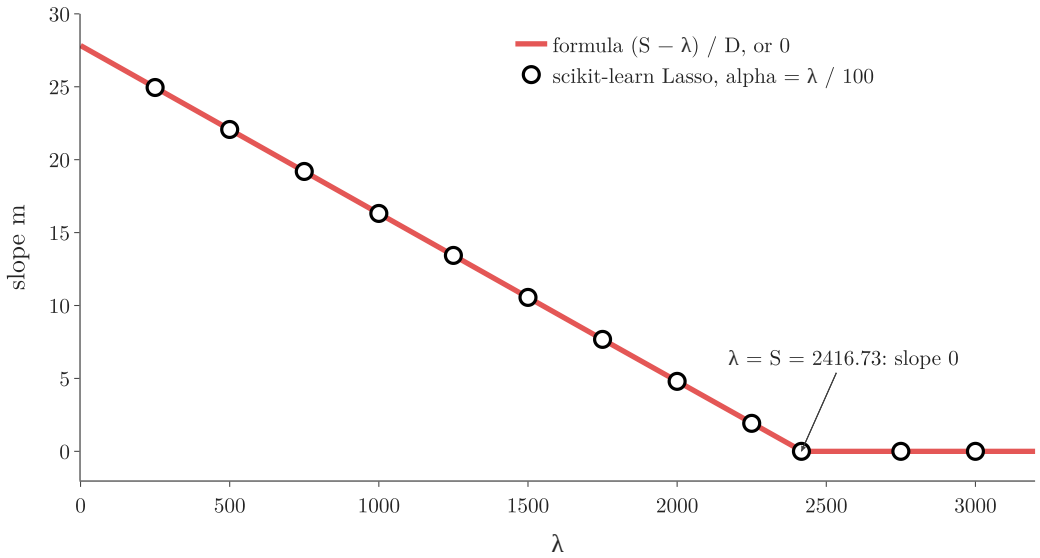


Figure 7: The one-feature Lasso slope on the 100-observation example: the formula (red line) and scikit-learn's Lasso (circles), for λ from 0 to 3000. The two agree to within 0.0001, and both are exactly 0 from $\lambda = S = 2416.73$ on.

scikit-learn's `Lasso` divides the squared error by $2n$, so its `alpha` equals λ/n. With $n = 100$, the slope reaches 0 at alpha $= 24.17$: the value found in one feature, the slope reaches exactly 0.

Python: The one-feature Lasso slope.

```python
def lasso_slope(S, D, lam):
    if S > lam:
        return (S - lam) / D
    if S < -lam:
        return (S + lam) / D
    return 0.0

lasso_slope(100, 50, 25)       # 1.5
lasso_slope(100, 50, 150)      # 0.0
```

7. Summary

	Ridge	Lasso
Slope, one feature	$S/(D+\lambda)$	$(S-\lambda)/D$, 0, or $(S+\lambda)/D$
Where λ appears	denominator	numerator
Large λ	slope small, never 0	slope exactly 0
Weak features	kept with small coefficients	dropped (dead zone)

- **Sparsity** (G-1849) means many coefficients exactly 0, so the features behind them drop out and the model is simpler and easier to read.
- The Lasso penalty puts a corner into the loss curve at **slope** (G-1823) 0; a large enough λ makes the corner the lowest point. The Ridge curve stays a smooth parabola, so its lowest point only slides towards 0.
- The **absolute value** (G-159) forces the Lasso formula into cases, because $|m|$ has no **derivative** (G-595) at 0; λ ends up subtracted from the numerator, so it can bring the slope to exactly 0.
- Once $\lambda \geq |S|$, the slope is 0 and stays there, because neither case formula then gives an answer with its own sign; so a feature only weakly linked to the **target** (G-1949) (small $|S|$) lands in the **dead zone** (G-552) and is dropped.
- In pictures: the loss ellipse usually first touches Lasso's diamond at a corner, where a coefficient is exactly 0, because the corners stick out towards the rings; Ridge's circle has no corners.
- This is the answer to the interview question: Lasso subtracts λ and Ridge divides by it, so only Lasso creates sparsity.

8. Sources

Built from

- CampusX, "Why Lasso Regression creates sparsity?", YouTube, https://www.youtube.com/watch?v=F
- StatQuest with Josh Starmer, "Ridge vs Lasso Regression, Visualized!!!", YouTube, https://www.youtube.com/watch?v=Xm2C_gTAl8c
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Other references

- **ESL:** Hastie, T., Tibshirani, R. and Friedman, J. *The Elements of Statistical Learning*, 2nd ed. Springer, 2009. Section 3.4.3 and Figure 3.11 (the diamond and the disk), and Section 3.8.6, p. 93.
- **scikit-learn docs:** `sklearn.linear_model.Lasso` (coordinate descent), scikit-learn 1.9.

9. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Sparsity (G-1849)	Having many coefficients exactly equal to 0.
Dead zone (G-552)	The range of S (how strongly the data pulls the slope away from 0) for which Lasso's best slope is exactly 0: inside it the penalty wins, which is how Lasso removes features.
Soft thresholding (G-1827)	Moving a value towards 0 by a fixed amount, and setting it to 0 if it would cross 0; this is how Lasso sets coefficients to exactly 0.
Coordinate descent (G-485)	An optimisation method that updates one coefficient at a time; used by scikit-learn's Lasso.
Slope (G-1823)	How much the output changes for one unit of change in the input; m in $y = mx + b$.
Lasso regression (G-1047)	Linear regression with a penalty on the sum of absolute coefficients (L1); it shrinks coefficients and can set some exactly to 0, which removes those features.
Ridge regression (G-1691)	Linear regression with a penalty on the sum of squared coefficients added to the loss; the penalty keeps the coefficients small, which reduces overfitting (L2 regularisation).
Sparse (G-1846)	Mostly made of zeros, such as a feature map whose values are almost all 0, or a model whose coefficients are mostly exactly 0.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Regularisation (G-1659)	Adding a penalty for large coefficients to the loss, so the model overfits less (lower variance).
Target (G-1949)	The output column we predict, such as the class; also called the label.
Residual sum of squares (sum of squared errors, SSE) (G-1684)	The squares of all the errors (residuals) added up; the quantity the best-fit line makes smallest, and a regression tree splits where the SSE of the two sides is smallest.
Observation (G-1374)	One record of a dataset: one row of the data table.
Intercept (G-960)	The line's value when the input is 0; b in $y = mx + b$.

Term	Meaning
Penalty term (G-1476)	The extra term added to the cost to discourage large weights.
Absolute value (G-159)	A number's size without its sign.
Derivative (G-595)	The slope of a function at a point: how fast the output changes as the input changes there. It is zero at the lowest point of a smooth curve, which is how training finds a minimum.
Differentiable (G-606)	Having a derivative (one clear slope) at a point, or at every point; the absolute value $\|m\|$ is not, at its corner $m = 0$. Gradient descent needs a differentiable loss.
Feature selection (G-768)	Keeping only the useful input columns and dropping the rest.
Constraint (G-456)	A condition the solution must satisfy; in SVM, $y_i(w^T x_i + b) \geq 1$ for every training point.
Feasible region (G-759)	All the points that obey every rule (constraint) of an optimisation problem; the answer must be one of them.
Ordinary least squares (OLS, least squares) (G-1406)	The way linear regression finds its line with one formula (a closed-form method): the line with the smallest sum of squared errors.
Contour plot (G-468)	A map of a surface seen from above, with lines joining points of equal height, so a 3D shape such as a loss bowl can be drawn and read on flat paper.